Valuing wildfire smoke related mortality benefits from climate mitigation

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Abstract

Human-induced climate change has increased wildfire risks, associated air pollution, and health damages in North America. Despite its large potential for damage, climate-induced wildfire smoke is rarely incorporated in estimates of the societal costs of climate change. We develop an integrated framework to estimate air pollution from climate-induced wildfire smoke (fine particulate matter, $PM_{2.5}$) and the associated mortality damage in the U.S. across different trajectories of greenhouse gas emissions and global mean surface temperature. Our framework accounts for fire-vegetation feedbacks by empirically estimating the effects of past fires on future burn probability. Under 3°C of global warming (relative to 1850-1900), we estimate an annual smoke-related mortality in the U.S. of 64,000 deaths (95%CI: 33,500-112,300; calculated using historical population) which increases the estimated mortality from smoke during 2011-2020 by 60%. Limiting global warming to 2°C reduces smoke-related mortality by 14% (8,900 deaths per year) relative to our estimate for 3°C. For every additional tonne of CO_2 emissions in 2025, we calculate a net present value of monetized damage (i.e., a partial social cost of carbon) of $11.2 (95%CI: -$1.1-$41.6; 2020USD) due to climate-induced wildfire smoke mortality in the U.S. Incorporating wildfire smoke damages into existing non-wildfire damage estimates increases the U.S. domestic social cost of carbon by 74% and thus substaintially increasing the expected benefits of greenhouse gas mitigation within the U.S.

Significance Statement

Wildfire smoke and associated health damages are increasing throughout the United States (US), partly driven by climate change. However, wildfire smoke is rarely incorporated in estimates of the societal costs of climate change, resulting in likely underestimation of the climate damages and expected benefits of climate mitigation. We estimate wildfire smoke pollution and mortality damages in the US across future climate scenarios. We find that limiting global warming to 1.5°C above the pre-industrial level can reduce smoke-related mortality in the US by 11,600 each year compared to 3°C. We estimate that every ton of CO_2 emitted leads to damage of $11.2 from smoke-related mortality in the US, which increases the US social cost of carbon by 74% relative to existing non-wildfire damage estimates.

Introduction

The global mean surface temperature (GMST) is currently 1.3-1.4°C above the pre-industrial level and is projected to reach 3°C of warming if current climate policies continue (*1*, *2*). Exceeding 2°C is an increasingly plausible outcome even if ambitious mid-century decarbonization goals are achieved (*3*). Failure to achieve the United Nation's Paris Agreement goals of limiting global warming to "well below" 2°C is likely to increase the frequency and intensity of many extreme weather events and their downstream effects on ecosystems and human society (*4–10*). Quantifying the frequency, intensity, and impacts of extreme climate events across global warming levels at local and regional scales is a key consideration for informing benefit-cost analysis of climate policies, highlighting local policy priorities, and designing climate adaptation plans. Valuation of potential damages from global warming (and thus potential benefits of climate mitigation) through metrics such as the social cost of carbon can inform international and domestic climate policy evaluation (*11*, *12*).

It is widely acknowledged that existing estimates of climate change impacts do not capture many potential pathways through which climate change influences human society (*13–17*). Among the many climate damages not fully captured in existing work, damage from increasingly frequent and severe wildfires is perhaps one of the most important, especially in fire-prone regions. Globally, wildfire frequency and intensity have increased substantially over the last two decades (*18–21*). In the United States (U.S.), wildfire burned area has quadrupled over the last forty years (*22*, *23*), driven in large part by human-induced climate change (*24–28*). Anthropogenic greenhouse gases (GHG) may account for over two-thirds of the increasing trend in Vapor Pressure Deficit (VPD) in the western U.S., an indicator of atmospheric moisture that can strongly predict wildfire risks (*29*). GHG emissions associated with the world's 88 largest carbon producers (scope 1 and scope 3 emissions) are estimated to account for 37% of the cumulative area burned from 1986-2021 in the western U.S. (*30*). In addition, population growth in the western U.S. has been fastest in areas where the vegetation is most sensitive to climate change, further amplifying risks (*31*).

As a result of increased wildfire activity, air pollution associated with wildfire smoke (specifically fine particulate matter, $PM_{2.5}$) has significantly increased in the U.S., undermining decades of progress on air quality (*32–37*). Exposure to wildfire smoke $PM_{2.5}$ negatively influences physical and mental health and increases mortality and morbidity burdens (*38–43*). Some evidence suggests that exposure to wildfire smoke $PM_{2.5}$ can be more damaging to human health than the same quantity of $PM_{2.5}$ originated from non-fire sources, although the relative toxicity remains an important question to be further explored (*44*, *45*). Under future climate change scenarios, wildfire smoke $PM_{2.5}$ concentrations are projected to increase in the U.S. (*46–51*), and lead to increased mortality, asthma incidents, respiratory hospitalizations, and other morbidity (*43*, *52–55*). While the projected health effects of future wildfire smoke $PM_{2.5}$ span a wide range, climate-induced wildfire smoke is estimated to generate health damages on the order of hundreds of billions of dollars annually when monetized using the value of mortality risk (*43*, *53*, *56*, *57*).

Despite the potential severity of damages from climate-induced wildfire smoke, leading estimates of the social costs of climate change often do not incorporate wildfire smoke damages. For example, the integrated assessment models that are widely used to calculate climate change damages, such as the Greenhouse Impact Value Estimator (GIVE) or Dynamic Integrated Climate-Economy (DICE) models, do not include wildfire smoke damage (*14*, *58*). To our knowledge, the Framework for Evaluating Damages and Impacts (FrEDI), developed by the U.S. Environmental Protection Agency, is the only climate impacts assessment model that explicitly considers wildfire smoke damage to human health (*56*). However, FrEDI only accounts for wildfires in the western U.S. and uses an empirical climate-fire relationship derived from historical data before 2013, which does not include recent extreme wildfire years (*54*). As a result, the estimated wildfire smoke damage is much smaller than more recent data-driven estimates (*43*). Given the substantial impacts of climate change on wildfire and the significant effects of wildfire smoke on human health, overlooking wildfire smoke damage could underestimate the cost of climate change in the U.S., which could in turn distort climate policy priorities.

In this study, we develop an integrated framework to estimate climate-induced wildfire smoke $PM_{2.5}$ concentrations and associated mortality damage in the contiguous U.S. across different levels of CO_2 emissions and GMST. We then use this framework to calculate the wildfire smoke-related mortality effects from climate mitigation through metrics of the social cost of carbon. While the focus is on smoke damages in the contiguous U.S., our approach also includes modelled changes in wildfire emissions in Canada and Mexico to capture important transboundary influences on U.S. smoke $PM_{2.5}$ and health effects. Our framework consists of four components (see Figure S1 for an overview). This framework builds on our prior work (*43*) which developed an empirical model that uses climate variables to predict wildfire smoke $PM_{2.5}$ concentrations under different climate conditions and an empirical mortality exposure-response function that is specific to wildfire smoke $PM_{2.5}$ (Methods).

The current modeling framework extends the model developed in Qiu et al. (2025) in three ways to quantify the wildfire smoke damages associated with global climate targets and carbon emissions. First, we combine the empirical model from Qiu et al. (2025) (*43*) with projections from 28 global climate models (GCMs) from the sixth phase of the Coupled Model Intercomparison Project (CMIP6) to project smoke $PM_{2.5}$ concentrations and mortality in the U.S. across a range of future climate scenarios from 2020 to 2099. This allows us to estimate "damage functions" that characterize how wildfire smoke $PM_{2.5}$ concentration and associated mortality vary as a function of GMST levels. Second, previous projections of future smoke $PM_{2.5}$ do not account for potential fire-fuel feedbacks, i.e., the fact that places that have recently burned are less likely to burn in the future. These feedbacks are particularly relevant to our end-of-century projection horizon. In this study, we empirically estimate how past fires can reduce future burn probability by using econometric methods and detailed historical fire data, focusing on western U.S. forests given the high current and projected level of fires in this area (and thus high relevance of potential feedbacks). We then incorporate these new estimates of fire-fuel feedbacks into our projections using a method similar to earlier work (see Methods and Figure S2) (*59*, *60*). Finally, we incorporate our wildfire smoke estimates into the GIVE integrated assessment model (*14*) to calculate the additional wildfire smoke mortality damages in the U.S. caused by an incremental tonne of CO_2 emissions (i.e., the social cost of carbon, or SC-CO_2). We further compare those wildfire damage estimates to U.S. non-wildfire climate damages previously quantified in the GIVE model.

Results

Figure 1 shows annual average wildfire smoke $PM_{2.5}$ concentrations over the contiguous U.S. across four different global warming levels (1.5, 2, 3, or 4°C warming above the 1850-1900 pre-industrial baseline). Wildfire smoke $PM_{2.5}$ concentrations increase substantially at all warming levels relative to the average for 2011-2020. Even under 1.5°C of warming, the annual average population-weighted smoke $PM_{2.5}$ in the U.S. reaches 0.84 $\mu g/m^3$, a 92% increase relative to the 2011-2020 level. Annual average population-weighted smoke $PM_{2.5}$ concentrations are projected to reach 1.2, 2.4, and 4.5 $\mu g/m^3$ in the U.S. under 2°C, 3°C, and 4°C warming, respectively. For comparison, the population-weighted smoke $PM_{2.5}$ was 0.90 $\mu g/m^3$ in 2020, one of the most extreme wildfire smoke years in recent U.S. history. At 3°C of warming, an average person in the U.S. is projected to be exposed to smoke pollution that is 2.5 times the 2020 level and 5.4 times the average smoke level in 2011-2020. Figure S3 shows the projected population-weighted smoke $PM_{2.5}$ concentration across different GMST levels. Projected increases of wildfire smoke $PM_{2.5}$ are largely driven by increases in wildfire emissions in the western U.S. and Canada due to lower soil moisture and higher ambient air temperature under higher warming levels (Figure S4). While the highest smoke concentrations are projected to occur in the western U.S., we also estimate substantial increases in smoke $PM_{2.5}$ concentrations across the country due to long-range transport of wildfire smoke.

As a result of the projected increase in smoke $PM_{2.5}$ concentrations under future warming, annual smoke-related excess deaths are estimated to increase to 64,000 (95%CI: 33,500 - 112,300) under the 3°C scenario under the historical population pattern. This is a 60% increase relative to 2011-2020 (Figure 1E and Table S1). Limiting global warming to 1.5°C or 2°C can reduce smoke-related mortality by 11,600 (18% reduction) or 8,900 (14% reduction), respectively, relative to 3°C warming. While smoke $PM_{2.5}$ concentrations are much higher in the western U.S. than in the eastern U.S., we estimate substantial smoke-related mortality in the eastern U.S. as well. Especially under historical and 1.5°C warming levels, we find that over 70% of smoke-related mortality comes from states that are east of 95° W, driven by their high population density and the significant effect of smoke $PM_{2.5}$ on mortality even at low annual mean concentrations (Figure S5). Nevertheless, smoke-related mortality increases more in the western U.S. due to higher climate-induced smoke $PM_{2.5}$ increases in the region. For example, limiting warming to 1.5°C can save 7,100 lives each year in California alone, relative to 3°C warming. We note that these results are calculated using the historical population density and distribution (held constant at the 2019 level) to isolate the influence of climate change on the smoke-related mortality burden. This likely underestimates the future smoke-related burden considering the projected increases in overall population and the aging trend (Table S2).

We find that incorporating fire-fuel feedbacks has important impacts on projected smoke $PM_{2.5}$ and mortality levels, especially under the high warming scenarios. Using historical fire data in western U.S. forests, we find that burning probability is 45%-65% lower in pixels that have recently burned, relative to pixels that have not burned before in western U.S. forests. This effect is found to last for seven years, after which the reduction in burning probability is no longer statistically significant (Figure S6, Table S3). Our results are similar to the estimates reported in (*59*), which found protective effects lasting for at least six years after low-intensity fires in California. When incorporated into future projections, effects of fire-fuel feedbacks are negligible in low-warming scenarios, because the probability of any individual pixel burning remains fairly low (Figure S7; See Methods for more details). However, the effects of feedbacks are much larger in high-warming scenarios. As a result, accounting for the fire-fuel feedback has little influence on estimated smoke-related mortality under 1.5°C or 2°C warming levels, but reduces the mortality estimate by 12% at 3°C and 24% at 4°C (Figure S8 and Table S1). Because of the important role of these feedbacks, all of our calculations (including Figure 1) incorporate fire-fuel feedbacks in projecting smoke $PM_{2.5}$ and resulting mortality.

To calculate a partial U.S. SC-CO_2 for wildfire smoke $PM_{2.5}$, we integrate our mortality damage estimates into the GIVE Integrated Assessment Model and monetize them using standard valuation approaches (Methods). We estimate damage functions using the mortality rate instead of absolute mortality estimate to allow flexible calculation for the damages across a variety of socioeconomic and demographic projections (as provided by the GIVE model; see Methods). For every million tonnes of CO_2 emitted in 2025, we estimate that wildfire smoke $PM_{2.5}$ associated with the additional warming causes 2 (95%CI: -0.19 - 6.96) excess deaths in the U.S. between 2025 and 2300. We estimate an average U.S. SC-CO_2 due to wildfire smoke mortality of $11.2 (95%CI: -$1.1-$41.6; 2020USD) per ton of CO_2 emitted in 2025 when accounting for the fire-fuel feedback (Figure 2). We assess the uncertainty in the U.S. SC-CO_2 using a Monte Carlo simulation of 10,000 runs, accounting for the uncertainty in baseline socioeconomic projections, the climate response to a marginal emission of CO_2, GCM-based projections of climate-driven wildfire smoke, and smoke-mortality exposure-response functions (Figure 2A). Our main estimate is calculated using a 2% near-term target dynamic discount rate following the central specification from (*14*), an emission pulse year of 2025, and a value of mortality risk of $10.05 million (2020USD). The smoke-related U.S. SC-CO_2 estimate is 68% larger if we do not account for the fire-fuel feedback in western U.S. (Figure S9). Table S4 shows estimates of U.S. SC-CO_2 under alternative discount rates and emission pulse years. Our U.S. SC-CO_2 estimate varies from $19.5 (2020USD) under a 1.5% discount rate to $4.6 (2020USD) under a 3% discount rate (Figure S10). We find largely similar results when estimating damage functions using alternative functional forms relating GMST and wildfire smoke damage, and alternative smoke-mortality exposure-response functions (Figure S11).

Our estimated U.S. SC-CO_2 due to climate-induced wildfire smoke is similar to the aggregate damage to the U.S. from all the other sectors previously considered in the GIVE model (Figure 2B). Using the same underlying socioeconomic projection pathways, climate model, and discount rate from GIVE, we calculate a U.S. SC-CO_2 of $15.2 accounting for all climate damages within the U.S. from the four sectors in the original GIVE model (temperature-related mortality, agriculture, energy, and sea level rise). Among these four sectors, temperature-related mortality accounts for

90% of the domestic climate change damages in the U.S. ($12.1) when not accounting for wildfire smoke damages. Accounting for wildfire smoke increases the estimated U.S. climate damages by 74%, which significantly increases the expected benefits from climate policies that reduce greenhouse gas emissions.

To extend these findings, we quantify their implications for cost-benefit analysis of the Inflation Reduction Act (IRA), which was projected to reduce CO_2 emissions by 193-985 million tonnes per year in 2030 in the U.S. (*61*). We estimate that for each year of these emissions savings, they are projected to avoid 1,101 cumulative deaths between 2025-2300 due solely to wildfire smoke $PM_{2.5}$ (range: 379-1,935, across the range of estimated emissions reductions). Using our U.S. SC-CO_2 estimates, we find that the IRA is projected to lead to annual health benefits of $6.9 billion (2020USD) (range: $2.4-$12.1 billion) from reduced climate-induced wildfire smoke mortality. These health benefits alone are equivalent to 20% of the estimated per-ton cost of reducing emissions associated with the IRA (*61*).

Discussion and Conclusions

Our research quantifies the costs of climate-induced wildfire smoke damage in the U.S. Our analysis extends an emerging literature that quantifies the health benefits of achieving global climate mitigation targets (such as the Paris Agreement), which has largely focused on the direct effects of temperature on mortality (*5*, *62*, *63*). Based on our estimates, reductions in wildfire smoke and related mortality are likely one of the most important benefits of GHG emission reductions in the U.S. As one measure of this importance, we find that incorporating wildfire smoke damage can increase the expected domestic benefits of U.S. climate mitigation by 74%. Given the magnitude of estimated damages, our results suggest a need to incorporate this channel into the climate impacts assessment tools used in policy evaluation.

Despite this substantial magnitude, ours is likely only a conservative estimate of the overall damage from wildfire, given possible additional morbidity effects of wildfire smoke (*42*), damages from other wildfire-driven pollutants (*64*, *65*), or direct damage from wildfires themselves (e.g., damaged structures (*66*)). In addition, our U.S. SC-CO_2 estimates only include damages occurring within the U.S. (although they include the trans-boundary effects on smoke in the U.S. due to changing fire risk in nearby countries). By contrast, recent U.S. benefit-cost analyses (*16*) have relied on SC-CO_2 estimates that account for damages occurring globally, reflecting the nature of

CO_2 as a global pollutant and consistent with recommendations by the U.S. National Academies (*13*). Nevertheless, U.S. SC-CO_2 values are useful for assessing the relative magnitudes of damages in different sectors and identifying local adaptation priorities, and may also be useful for decision-makers who find domestic climate benefits to be particularly relevant (*67*).

While prior studies often calculate the total societal impact of climate change by adding individual pathways together, there is potential overlap or interaction between these pathways. In our case, one important concern is whether smoke-related mortality is separable from temperature-related mortality, given that wildfire smoke is correlated with ambient temperature. To address this concern, we flexibly control for temperature when estimating the exposure-response function between wildfire smoke $PM_{2.5}$ and mortality (Methods). Thus, the smoke-mortality exposure-response function isolates the effects of elevated smoke $PM_{2.5}$ concentrations on mortality from the effects of temperature. For temperature-mortality in the GIVE model, the calculation is based on a temperature-mortality relationship drawn from a global meta-analysis, which incorporated many studies conducted in regions where smoke is not a particular issue of concern (*68*). Therefore, we believe it is appropriate to add estimates of smoke-mortality and temperature-mortality to quantify their total damage. In addition, previous studies have also suggested that the joint effect of heat and smoke on human health may be worse than the sum of the individual effects alone (*69*) and thus our estimate of total climate change effects (after including smoke) may be viewed as conservative. A careful analysis of temperature effects on mortality while controlling for smoke $PM_{2.5}$ will be important for a comprehensive understanding of the separate and joint mortality impacts of temperature and wildfire smoke.

In this study, we directly quantify the effect of past fires on future burning probability in western U.S. forests, which has been identified as the leading channel of fire-fuel feedback (*60*), and incorporate this feedback in our fire projection models. We find that incorporating our empirical estimate of the fire-fuel feedback has little influence on projected fire emissions in lower warming scenarios, which include near-term settings in which substantial warming has yet to occur, but can significantly reduce projected fire and smoke in the long term (i.e., by 30-40% in 2100, relative to estimates without incorporating these feedbacks). Given the long time horizons for SC-CO_2 calculation, we find that our smoke-related U.S. SC-CO_2 estimate will be 68% larger without accounting for the fire-fuel feedback (Figure S9). Nevertheless, this feedback is unlikely to reverse the increasing trend in fire emissions, smoke, and mortality under future climate. Our estimates of the feedback and the feedback's influence on fire emissions are comparable to prior studies that used process-based dynamic vegetation models or statistical approaches (*59*, *60*, *70*). Nevertheless, we acknowledge that our fire-fuel feedback estimates are uncertain, particularly for high levels of warming that fall outside of our empirical baseline. Opportunities for improvement include leveraging process-based fire-vegetation models, longer records of fire history, and more advanced causal inference approaches. Our estimates also do not capture the effects of past fires on future fire severity, intensity, and shifts in vegetation composition or species, which may influence projected fire emissions, smoke, and mortality.

Given the increasing wildfire smoke and the health damages, different adaptation measures that may reduce future smoke burden and associated health damage are increasingly discussed and implemented. Emerging research has evaluated various adaptation measures at a local scale, including the use of prescribed fires to reduce future wildfire emissions (*71*), as well as the use of indoor air filters to reduce indoor air pollution (*72*). However, to our knowledge, there is no known evidence that climate-smoke or smoke-mortality relationships are currently diminishing in a way consistent with adaptation. For example, unlike some other climate change damages, we do not find a protective effect of income on the health impacts of wildfire smoke pollution (Figure S12), or heterogeneity in the smoke-mortality relationship across baseline smoke levels (Figure S13). This pattern is consistent with the current evidence that relying on individuals to self-protect appears inadequate (*73*). However, novel adaptation strategies could occur in the future. If the cost of their design and implementation is less than the monetized benefits from reduced smoke impacts, then our estimates will overstate total damages. Given the large uncertainty in the speed and scale with which they could be adopted, our modeling framework does not explicitly consider adaptation measures that have yet to occur. Identifying and implementing cost-effective strategies to reduce wildfire risks and associated smoke impacts amid a warming climate is an urgent priority for both research and policy.

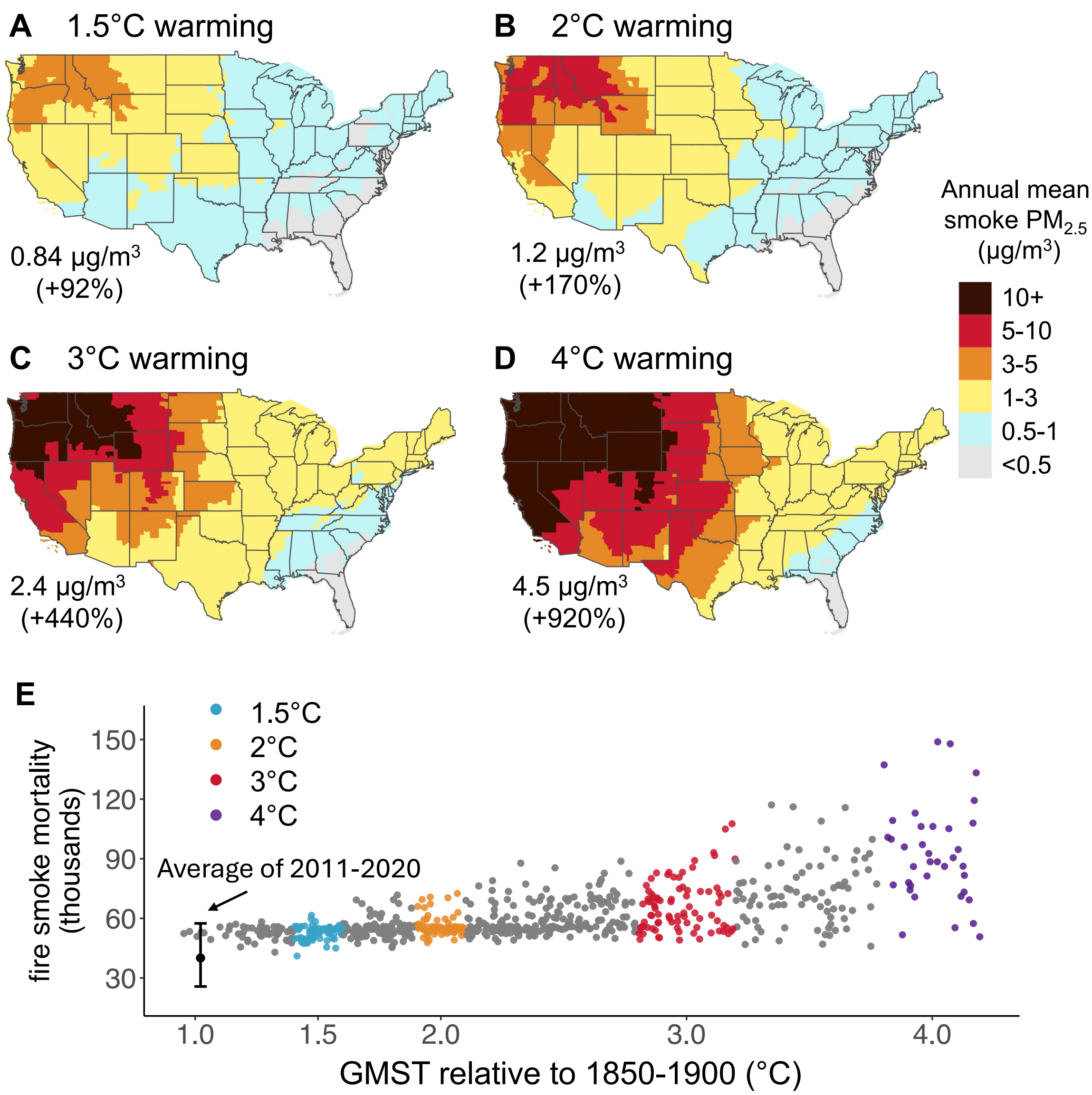

Figure 1: **Projected wildfire smoke $PM_{2.5}$ concentrations and estimated premature mortality across different global warming levels.** Panels A-D show the annual mean smoke $PM_{2.5}$ concentration at 1.5°C, 2°C, 3°C, and 4°C warming relative to 1850–1900. Each panel shows the annual mean smoke $PM_{2.5}$ concentrations (median of all climate models). The text in the figure shows the population-weighted annual mean smoke $PM_{2.5}$ concentrations across the contiguous U.S. The percentage in the parentheses shows the relative increase in population-weighted mean smoke $PM_{2.5}$ relative to 2011–2020. Panel E shows the annual estimated smoke-related mortality across different global mean surface temperatures (GMST). Population distribution and density are fixed at the 2019 level. Each dot shows the estimated annual mortality from an individual GCM, averaged over 20 years. Colored dots show those used to calculate 1.5°C, 2°C, 3°C, and 4°C warming scenarios. Estimates incorporate empirical estimates of feedback from fire activity on future burn probability.

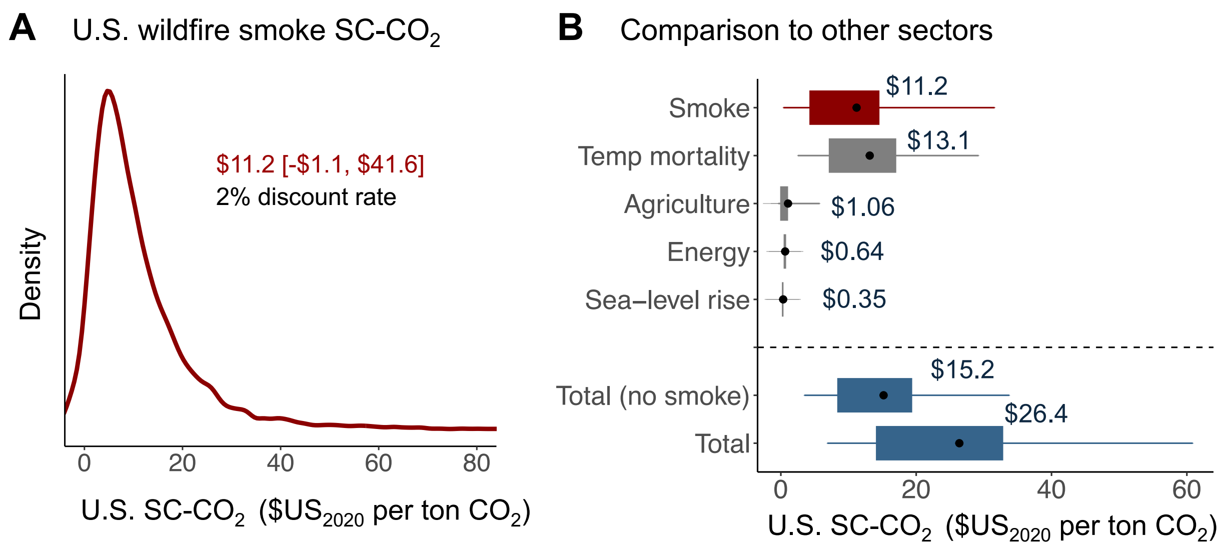

Figure 2: **U.S. social cost of carbon due to wildfire smoke mortality.** Panel A shows the distribution of the estimated U.S. SC-CO_2 values from 10,000 Monte-Carlo simulations using a 2% near-term discount rate and emission year 2025. The mean value and 95% confidence interval (2.5th and 97.5th percentile) are shown in the text. Panel B shows the U.S. SC-CO_2 due to wildfire smoke mortality in comparison to the other four sectors included in the GIVE model. The portion below the dashed line shows the total U.S. SC-CO_2 with and without wildfire smoke damage. All U.S. SC-CO_2 shown here represent damages only occurring within the U.S. On each box-and-whisker, the dot shows the mean value, the rectangle shows the 25th and 75th percentiles, and the line shows the 10th and 90th percentiles. Units of U.S. SC-CO_2 are U.S. dollars in the year 2020 per ton of CO_2.

Materials and Methods

Climate change impacts on wildfire smoke $PM_{2.5}$

We use the empirical model developed in Qiu et al., 2025 to characterize the relationship between climate variables and wildfire smoke $PM_{2.5}$ concentrations in the contiguous U.S. (48 states and the District of Columbia). A high-level description of this model is provided below. Model performances and further method details can be found in (*43*). Qiu et al. developed an ensemble of machine learning and statistical models to predict wildfire emissions using climate and land use variables ("climate-fire model"), and a statistical regression model to predict wildfire smoke $PM_{2.5}$ concentrations using wildfire emissions, accounting for the distance between fires and smoke locations and wind directions ("fire-smoke model"). This model can characterize how climate variables influence the wildfire emissions in North America (including the contiguous U.S., Canada, and Mexico), and annual mean wildfire smoke $PM_{2.5}$ concentrations in the contiguous U.S.

For the climate-fire model, Qiu et al. developed an ensemble of statistical and machine learning models using both climate and landuse variables to predict wildfire emissions. Separate climate-fire models were constructed for each of the five regions (western U.S., southeastern U.S., northeastern U.S., Canada-Alaska, and Mexico; region definitions can be found in Figure S4). The outcome variable is monthly or annual dry matter (DM) emission, which represents the estimated biomass consumed in the burning process, derived from the fourth version of the Global Fire Emissions Database with small fires (GFED4s) from 2001-2021 (*74*). The input features include 2m air temperature, precipitation, relative humidity, soil moisture, vapor pressure deficit (VPD), wind speed, runoff, and land use types. The climate-fire models achieve high prediction performance in the western and southeastern US, Canada-Alaska, and Mexico, with correlation coefficients of 0.79-0.95 in out-of-sample evaluations, demonstrating higher performance relative to prior statistical methods to model climate impacts on burned area (more details can be found in (*43*)).

For the fire-smoke model, Qiu et al. estimated how variations in wildfire emissions influence wildfire smoke $PM_{2.5}$ concentrations in the contiguous U.S. (at 10 km resolution). Separate regression coefficients between emissions and smoke concentrations are estimated for different wind directions and different distances from fires. The combined climate-fire and fire-smoke models are able to predict historical smoke $PM_{2.5}$ concentrations with a correlation coefficient of 0.72 at the grid cell level in out-of-sample predictions (more details can be found in (*43*)).

Projecting wildfire emissions and smoke $PM_{2.5}$ concentration

We combine the climate-fire-smoke model developed in Qiu et al. with 784 total realizations from 28 global climate models (GCMs) archived in Phase 6 of the Coupled Model Intercomparison Project (CMIP6). For each GCM, we use the simulated climate variables averaged over seven 20-year periods (2020-2039, 2030-2049, 2040-2059, 2050-2069, 2060-2079, 2070-2089, 2080-2099) from four scenarios (SSP1-2.6, SSP2-4.5, SSP3-7.0, and SSP5-8.5) featured by Intergovernmental Panel on Climate Change (IPCC) (*75*). For each of the 784 GCM realizations, we predict the annual wildfire DM emissions at different spatial resolutions.

Because weather and climate variables are more predictive of wildfire emissions at aggregated spatial scales rather than at the individual grid level (*76*), the climate-fire model predicts wildfire emissions at aggregated regional scales. To further estimate smoke $PM_{2.5}$ concentrations at the grid cell level, we use a stochastic downscaling method that was developed in (*43*) to estimate emissions at the grid cell level (0.25 degree). This stochastic downscaling method estimates fire emission for each grid cell using a random draw from a gamma distribution while ensuring the sum across all grids equals the aggregated regional predictions (more details can be found in (*43*)). The downscaling approach allows us to capture potential fire emissions from grid cells that have not burned before and assess the uncertainty in the downstream smoke and health consequences due to fire locations, given the same regional fire emission predictions. We find that averaging across all downscaling estimates produces smoke estimates generally consistent with the historical smoke estimates.

We then combine the downscaled predicted DM emissions at GFED4s grid cell level with the empirical relationship we established between smoke $PM_{2.5}$ and DM emissions to calculate smoke $PM_{2.5}$ at the resolution of 10 km. When calculating the smoke $PM_{2.5}$ in future scenarios, the wind direction is held constant at the average conditions in the historical period due to the large uncertainty in the future projections of wind patterns. We further calculate the difference between estimated smoke in any future year and the average *estimated* smoke between 2011-2020. The delta difference is then added to the average *observed* smoke $PM_{2.5}$ concentration between 2011-2020 to obtain the final smoke predictions for each grid cell in future years.

Incorporating fire-fuel feedback

Previous studies investigating the fire-vegetation-climate feedback have found that, by altering vegetation biomass, structure, and species, past fires can influence future fire size, intensity, severity, and emissions (*59*, *70*, *77*, *78*). Prior studies have also found that this feedback can differ significantly across different ecosystems and vegetation types (*79*). Our objective here is to assess the potential influence of this feedback on future smoke, mortality, and the U.S. SC-CO_2 values, rather than comprehensively evaluate all potential fire-vegetation-climate feedbacks. Thus, we empirically estimate this feedback using a method that is consistent with our overall climate-fire-smoke framework.

We quantify the influence of fire-fuel feedback on future fire emissions in forested areas of the western U.S., given high projected fire emissions in this region under future climate scenarios. We adopt a similar framework as (*60*) and focus on the effects of past fires on future burning probability, an important channel of fire-fuel feedback identified in (*60*). We use gridded fire radiative power (FRP) data reported by the Fire Information for Resources Management System (FIRMS) in the states of California, Oregon, and Washington, from 2000 to 2023. FRP is the radiative component of the fire energy release, and is calculated using imagery from MODIS Terra and Aqua. We use the nominal confidence ($> 30\%$) threshold to remove possible false-positive burning pixels. We create a dummy variable "fire" to indicate if a pixel has burned or not (fire $= 1$ if FRP > 0) and aggregate the data to 1km spatial scale and annual temporal scale. To reduce the potential influence of multiple fire events on our estimation, we only include pixels that have never burned or burned up to twice during this period (see below for sensitivity analyses). Of these, 4% of pixels have burned twice, 17% burned once, and 79% never burned during our observed study period. We further create a variable "first fire year" indicating the first fire year of each pixel in our studied period, with the value set to 10,000 for pixels that have never burned in our dataset.

Our statistical method follows an approach from (*59*). The idealized natural experiment is to compare a pixel that burned in year t with a pixel that is otherwise similar but has not burned in year t, and compare if their burning probability differs afterward (year t is referred to as the "focal year"). To consider all possible comparisons during our study period, we recompile and stack our data by focal years (from 2000-2022) and define "treatment groups" accordingly. For example, for the focal year of 2005, we include all pixels whose first fire year ≥ 2005 (thus including pixels that have never burned). We assign all pixels that burned in 2005 to the treatment group and the rest to the control group (Figure S2). We quantify the influences of historical fire activity on current burning probability using the following regression equation:

$$\text{fire}_{iy} = \sum_{t=1,...} \beta_t \times \text{treat}_{iy} \times \mathbf{1}(\text{relative year}{=}t)_{iy} + \alpha_c + \delta_y + \epsilon_{iy} \tag{1}$$

where fire_{iy} is a dummy variable that denotes whether pixel i was burned in year y; treat_{iy} equals one for pixels/years in the treatment group, and zero otherwise; *relative year* is the difference between the current year and the focal year and only appears in the interaction terms; and β_t are the main parameters of interest which quantify changes in burning probability after t years of a prior fire, relative to the control group. In our main specification, we include county (α_i) and year (δ_y) fixed effects (i.e., county- or year-specific intercepts) to capture differences across grids in average fire risk as well as overall trends in burn probability over time. We include all data that have focal years >2005 to ensure that prior fire histories are properly considered. Fires prior to 2000 are not observed by us and thus could influence the estimation on samples of the earlier focal years. Nevertheless, our results are robust to alternative specifications and focal year selections (Table S3 and Figure S6).

To calculate the influence of fire-fuel feedback on projected fire emissions, we first calculate the relative risk of burning by normalizing β_t by the baseline burning probability in the control groups (denoted as rr_t, shown in Table S3). For pixels that are projected to have burned more than once during the past seven years, we only consider the most recent fire. We further aggregate this relationship to the resolution of our projected emissions (0.25 degrees). Thus, the projected emissions with fire-fuel feedback incorporated are calculated as:

$$\text{Emis}_{iy} = \text{Emis}_{iy}^{baseline} \times (1 + \sum_{t=1,..,7} (rr_t \times \text{Fraction of cells burned in } y\text{-}t)) \tag{2}$$

where $\text{Emis}_{iy}^{baseline}$ is the projected emissions without accounting for this feedback, and *Fraction of cells burned in year* $y - t$ represents the percentage of projected burned area out of the total max eligible burned area. We only consider the feedback up to 7 years after the initial burning, given that our empirically derived feedback becomes statistically insignificant after 7 years.

Two parameters affect the “strength” of the feedback. The first is the regression estimate of changes in relative risk of burning subsequent to initial burning, as described above. The second is the fraction of cells that are estimated to burn in each period, to which the relative risk estimate will be applied. If this fraction is one, then all pixels in this grid cell have burned in the past and are subject to adjustment due to the fire-fuel feedback. However, if this fraction is closer to zero, the influence of fire-fuel feedback is much smaller, because the number of pixels that are projected to burn more than once is much lower.

In our main analysis, we only include pixels that burned 0-2 times, and the control group includes pixels that did not burn in the focal year, which includes pixels that never burned and pixels that burned after the focal year. Thus, our sample possibly includes pixels that have minimal fire risks in the control group, leading to potential downward bias in the estimated fire-fuel feedback. We conduct a sensitivity analysis re-estimating our model with an alternative sample by including pixels that burned more than twice and only pixels with "moderate", "high", or "very high" fire risks as determined by the CALFIRE Fire Hazard Severity Zones (*80*). The alternative sample includes 50432 pixels compared to 291906 pixels in the original sample. As shown in Figure S14, we find that the estimated fire-fuel feedback is smaller using the more restricted sample compared to the full sample, yet the overall pattern is similar. Using these new coefficients, we find that the simulated fire emissions are slightly larger than the estimated emissions using the original feedback coefficients, yet still lower than the baseline emissions (i.e., no feedback is assumed). Thus, our main estimate should be viewed as a conservative estimate for the future emissions and smoke-related damage.

Wildfire smoke $PM_{2.5}$ effects on mortality

We calculate all-cause mortality associated with wildfire smoke exposure using a two-part exposure-response function (Figure S15). For projected smoke $PM_{2.5}$ concentrations that are below the historical maximum level (annual mean concentration $< 6\mu g/m^3$), we use the exposure-response function developed in (*43*). Qiu et al. used a Poisson distributed lag model to estimate the effects of smoke $PM_{2.5}$ on all-cause mortality rate in the contiguous U.S. from 2006-2019. They found that smoke exposure not only increases mortality rates in the same year but also in the following years. For example, exposure to annual mean smoke concentration between 0.75-1 $\mu g/m^3$ increases the mortality rate by 1.5% in the same year and by 2.7% when accounting for deaths occurring in the following year. In this study, we use their main estimates which report cumulative mortality during the smoke year and the following year ("lag 0-1"). Given the annual-level projections of wildfire smoke, we choose to use this exposure-response function that is based on the annual mean smoke $PM_{2.5}$ concentration. Qiu et al. found that using cumulative short-term daily or monthly extremes to estimate mortality impacts yields similar estimates to using annual averages, suggesting the mortality effects of annual smoke concentration are driven by individual extreme events (*43*).

For projected smoke $PM_{2.5}$ concentrations that exceed the historical maximum (i.e. annual mean concentration $> 6\ \mu\text{g/m}^3$), we use an exposure-response function derived from a recent meta-analysis of mortality effects of all-source total $PM_{2.5}$ (*81*). We use this exposure-response function instead of the smoke-specific exposure-response function because the binned exposure-response function reported in (*43*) assumes the same mortality effects for concentrations that exceed the historical maximum (e.g., the estimated mortality will be the same for populations that are exposed to annual mean concentrations of 6 $\mu\text{g/m}^3$ or 50 $\mu\text{g/m}^3$). Analyzing 16 prior studies that focused on the North American population, Pope et al. estimated a 0.8% increase in mortality rates per 1 $\mu\text{g/m}^3$ of all-source $PM_{2.5}$ (*81*). Their estimate is lower than the mortality effect derived from the smoke-specific exposure-response function at low annual concentrations, but comparable at the high end of the historical maximum annual mean smoke $PM_{2.5}$ concentration (Figure S15). Figure S16 shows the relationship between the population-weighted smoke $PM_{2.5}$ and smoke-related mortality. There is a clear two-phase pattern where the smoke-related mortality does not increase as fast when the projected smoke level is low – due to lower estimated mortality risk in the range of 1-4 $\mu\text{g/m}^3$, a pattern observed in prior literature (*43*, *82*). In a sensitivity analysis, we further estimate the mortality-GMST damage functions using the exposure-response function from Pope et al. alone (Figure S11).

Linking wildfire smoke damages to global mean surface temperature

For each climate model, we first calculate the global mean surface temperature (GMST) using area-weighted surface temperature across all grid cells globally. We then calculate global warming levels using the difference between future GMST (20-year mean) and the average GMST during 1850-1900 following practices from IPCC (*75*). The 784 climate simulations span a wide range of possible warming levels, from 0.95°C to 6.93°C warming above the 1850-1900 levels. Out of the 784 climate simulations, we select 70 simulations with projected global warming level of 1.4-1.6°C as the "1.5°C warming scenarios", 82 simulations with warming level of 1.9-2.1°C as the "2°C warming scenarios", 86 simulations with warming level of 2.8-3.2°C as the "3°C warming scenarios", and 37 simulations with warming level of 3.8-4.2°C as the "4°C warming scenarios". We then estimate damage functions to characterize the relationship between wildfire smoke-related mortality rate in the U.S. (unit: deaths per million) and GMST. We use a quadratic model in our main analysis, and also find similar results using a 4th order polynomial model (Figure S11). We construct damage functions using mortality rate instead of absolute mortality estimate to allow flexible calculation for the damages across a variety of socioeconomic and demographic projections (as provided by the GIVE model; see section below). We assess the uncertainty of the damage functions with 10,000 bootstrap runs, which account for the uncertainty from both the climate models and the smoke-mortality exposure-response function (Figure S17).

Calculating the social cost of carbon

We calculate the partial U.S. SC-CO_2 for wildfire smoke by incorporating the damage functions into the open-source Greenhouse Impact Value Estimator (GIVE) Integrated Assessment Model (*14*). The GIVE model uses methods recommended by the National Academies (*13*) and adopted by the U.S. government in recent benefit-cost analyses (*16*). Specifically, it leverages probabilistic estimates of future emissions and socioeconomic trajectories, a simple climate model to calculate the temperature response to CO_2 emissions, and an approach to discounting that directly incorporates uncertainty about future economic growth.

Our core modification to GIVE is to add a component expressing deaths from wildfire smoke $PM_{2.5}$ in the U.S. as a function of GMST, using the smoke-related damage functions described above. This component complements existing sectors that are included in the original GIVE model: temperature-related mortality, agriculture, energy, and sea level rise. As our U.S. SC-CO_2 calculations use the same underlying socioeconomic projection pathways, climate model, and discount rate, the resulting U.S. SC-CO_2 from wildfire smoke can be directly added to or compared to the other sectoral U.S. SC-CO_2 estimates already in GIVE.

We follow the standard experimental design of U.S. SC-CO_2 calculations in running two simulations with GIVE: one using baseline emissions and socioeconomic trajectories, and one with an additional pulse of 0.1 million tonnes of CO_2 in a given year (2025, in our main estimates). Temperature trajectories and the associated damages are run to 2300 in both simulations, and the difference in damages between the simulation with the additional CO_2 and the baseline simulation is the impact of the marginal emission of CO_2. This difference is converted to net present value terms by discounting and summing over the period between the emissions pulse and 2300. The smoke-GMST relationship is estimated using data from 2020-2099 and is assumed to be valid during 2100-2300. We note that while the smoke-GMST relationship is estimated using projections from 2020-2099, the different climate models cover a wide range of temperature projections that largely overlaps with the range of temperature projections from the GIVE model. Following previous work with GIVE (*14*), we use a Ramsey-like discount rate whereby the discount factor is a function of two parameters (the pure rate of time preference and the rate of global economic growth), with parameters chosen to calibrate the near-term target discount rate to 2% over the next 10 years. We use a near-term target discount rate of 2% for our main estimate, and report estimates across alternative near-term rates from 1.5% to 3% (Figure S10). We assess uncertainty in our U.S. SC-CO_2 estimates with 10,000 Monte Carlo simulations (*14*). These simulations sample uncertainty in the socioeconomic and emissions trajectories, climate model parameters, sea level model parameters, and damage functions.

Because the socioeconomic projections provided by the GIVE model do not include population projections by age distributions, we use the all-age exposure-response function in (*43*) to estimate smoke-related mortality rate across different warming levels. This allows a direct comparison between our estimates and other damage channels already considered in GIVE. However, given the projected shifts in the age distribution (e.g., an overall aging of the society), our approach could underestimate the future smoke mortality and damages. Table S2 shows the estimated smoke-related mortality across different warming levels and population data.

When we quantify the smoke-related benefits of the Inflation Reduction Act (IRA), we use the U.S. SC-CO_2 based on an emissions pulse year of 2030 instead of 2025, because previous work on the IRA reported estimated emissions reductions in 2030 (*61*). Estimated emissions savings and mitigation cost are derived from Bistline et al. 2030 (*61*). Values of our U.S. SC-CO_2 estimates that correspond to 2030 emissions pulses can be found in Table S4.

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Author contributions

MQ, CWC, and MB designed the research. MQ led the analysis on fire, smoke, and mortality projection with input from IHM. IHM provided fire data. CWC led the calculation of social cost of carbon with input from LR. MQ led the manuscript writing with input from all authors. All authors reviewed and approved the manuscript.

Competing interests

The authors declare no competing interests.

Data and materials availability

All data and code will be made available on a public repo upon publication. The existing GIVE model is publicly available at https://github.com/rffscghg/MimiGIVE.jl. Our modifications to GIVE will be fully open-source and accessible to the community upon publication.

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SI tables and figures

Table S1: Estimated annual excess deaths due to wildfire smoke $PM_{2.5}$. For the historical period, the table shows average annual excess deaths due to smoke $PM_{2.5}$ exposure during 2011-2020. For future climate scenarios, the table shows average annual excess deaths due to smoke $PM_{2.5}$ exposure across different warming levels (median across 28 GCMs). Values in the parentheses show the 95% confidence interval across GCMs and uncertainty in the mortality exposure-response functions.

climate scenario	fire-fuel feedback	deaths due to smoke
1.5 deg	no feedback	52,900 (33,800 – 77,900)
	with feedback	52,500 (33,700 – 76,200)
2 deg	no feedback	56,300 (34,400 – 88,900)
	with feedback	55,100 (34,000 – 83,200)
3 deg	no feedback	69,300 (34,800 – 140,700)
	with feedback	64,000 (33,500 – 112,300)
4 deg	no feedback	112,400 (42,900 – 247,600)
	with feedback	88,800 (38,000 – 168,500)
2011-2020		40,100 (25,700-57,500)

Table S2: Estimated smoke-related mortality across different warming levels and population data. The table shows results calculated using the historical population in 2019 and projected population data in 2050 and 2100, respectively. Projected population and mortality rate are derived from the U.S. Census (*83*). Here the estimated mortality is calculated using the age-specific exposure-response function from (*43*), for the below-65 and 65+ groups, respectively.

population	**baseline death rate**	**1.5 degree**	**2 degree**	**3 degree**	**4 degree**
historical 2019	historical	52468 (33654, 76195)	55137 (34036, 83208)	64045 (33514, 112322)	88780 (37964, 168487)
future 2050	historical	56499 (36240, 82049)	59374 (36651, 89601)	68966 (36089, 120953)	95602 (40881, 181433)
future 2050	future 2050	72966 (46802, 105963)	76679 (47334, 115716)	89066 (46608, 156205)	123465 (52795, 234311)
future 2100	historical	57270 (36734, 83168)	60184 (37151, 90823)	69907 (36582, 122603)	96906 (41438, 183907)
future 2100	future 2100	79803 (51188, 115891)	83864 (51769, 126559)	97412 (50975, 170841)	135034 (57742, 256267)

Table S3: Estimated effects of fires on future burning probability in western U.S. forest. The table shows the relative change in burning probability X years after a fire compared to a pixel that had not burned previously in our sample period. "Main" shows our main estimate reported in the main text using the county and year fixed effects and excluding data with focal years $<$ 2005. "All years" shows results estimated using all data across all focal years from 2000-2022. "Focal year $\geq$ 2010" shows results estimated data with only focal years $\geq$ 2010. "county $\wedge$ year FE" shows results estimated using county-year fixed effects. "L3+year FE" shows results estimated using level-III ecoregion and year fixed effects. Significance level: $***$ p<0.01, $**$ p<0.05, $*$ p<0.1.

Years after fire	Main	All years	Focal year >= 2010	county^year FE	L3 + year FE
1	-0.451 (0.291)	-0.311 (0.253)	-0.583 (0.371)	-0.224 (0.285)	-0.113 (0.210)
2	-0.446 (0.140)***	-0.358 (0.113)***	-0.494 (0.170)***	-0.055 (0.097)	-0.094 (0.097)
3	-0.608 (0.185)***	-0.483 (0.160)***	-0.624 (0.207)***	-0.246 (0.118)**	-0.321 (0.164)**
4	-0.650 (0.162)***	-0.625 (0.133)***	-0.586 (0.165)***	-0.404 (0.125)***	-0.401 (0.145)***
5	-0.658 (0.138)***	-0.631 (0.117)***	-0.612 (0.153)***	-0.382 (0.144)***	-0.415 (0.124)***
6	-0.598 (0.179)***	-0.603 (0.153)***	-0.484 (0.212)**	-0.522 (0.168)***	-0.379 (0.162)***
7	-0.444 (0.142)***	-0.451 (0.122)***	-0.311 (0.165)*	-0.263 (0.128)**	-0.224 (0.119)*
8	0.294 (0.458)	0.177 (0.394)	0.046 (0.332)	0.342 (0.318)	0.516 (0.468)
9	0.331 (0.432)	0.182 (0.390)	0.571 (0.886)	-0.010 (0.351)	0.526 (0.453)
10	-0.275 (0.154)*	-0.328 (0.138)***	-0.182 (0.156)	-0.203 (0.179)	-0.088 (0.144)
11	-0.235 (0.129)*	-0.267 (0.105)***	-0.206 (0.205)	-0.106 (0.177)	-0.013 (0.124)
12	0.339 (0.387)	0.056 (0.318)	-0.521 (0.217)***	-0.170 (0.202)	0.551 (0.402)
13	0.484 (0.809)	0.284 (0.584)	-0.160 (0.415)	0.389 (0.647)	0.703 (0.829)

Table S4: Estimated U.S. SC-CO_2 across a range of discount rates and emission years (unit: U.S. dollar in year 2020). Our main estimate is calculated using a 2% discount rate and emission year of 2025 (shown in bold) with fire-fuel feedback.

discount rate	emission year	with fire-fuel feedback	no feedback
1.5% Ramsey	2020	18.1 (-1.8 - 67.0)	30.7 (-1.2 - 116.4)
1.5% Ramsey	2025	19.5 (-1.8 - 72.2)	33.2 (-1.3 - 126.6)
1.5% Ramsey	2030	21.1 (-1.9 - 79.3)	35.9 (-1.3 - 138.4)
2.0% Ramsey	2020	10.2 (-1.1 - 37.4)	17.1 (-0.6 - 65.6)
2.0% Ramsey	**2025**	**11.2 (-1.1 - 41.6)**	18.8 (-0.5 - 72.2)
2.0% Ramsey	2030	12.3 (-1.1 - 46.2)	20.7 (-0.5 - 79.6)
2.5% Ramsey	2020	6.2 (-0.8 - 23.2)	10.2 (-0.5 - 39.4)
2.5% Ramsey	2025	6.9 (-0.8 - 26.2)	11.5 (-0.4 - 44.4)
2.5% Ramsey	2030	7.7 (-0.8 - 28.7)	12.9 (-0.3 - 48.7)
3.0% Ramsey	2020	4.0 (-0.6 - 14.9)	6.5 (-0.4 - 24.8)
3.0% Ramsey	2025	4.6 (-0.7 - 16.8)	7.5 (-0.3 - 28.3)
3.0% Ramsey	2030	5.2 (-0.6 - 19.0)	8.5 (-0.3 - 31.8)

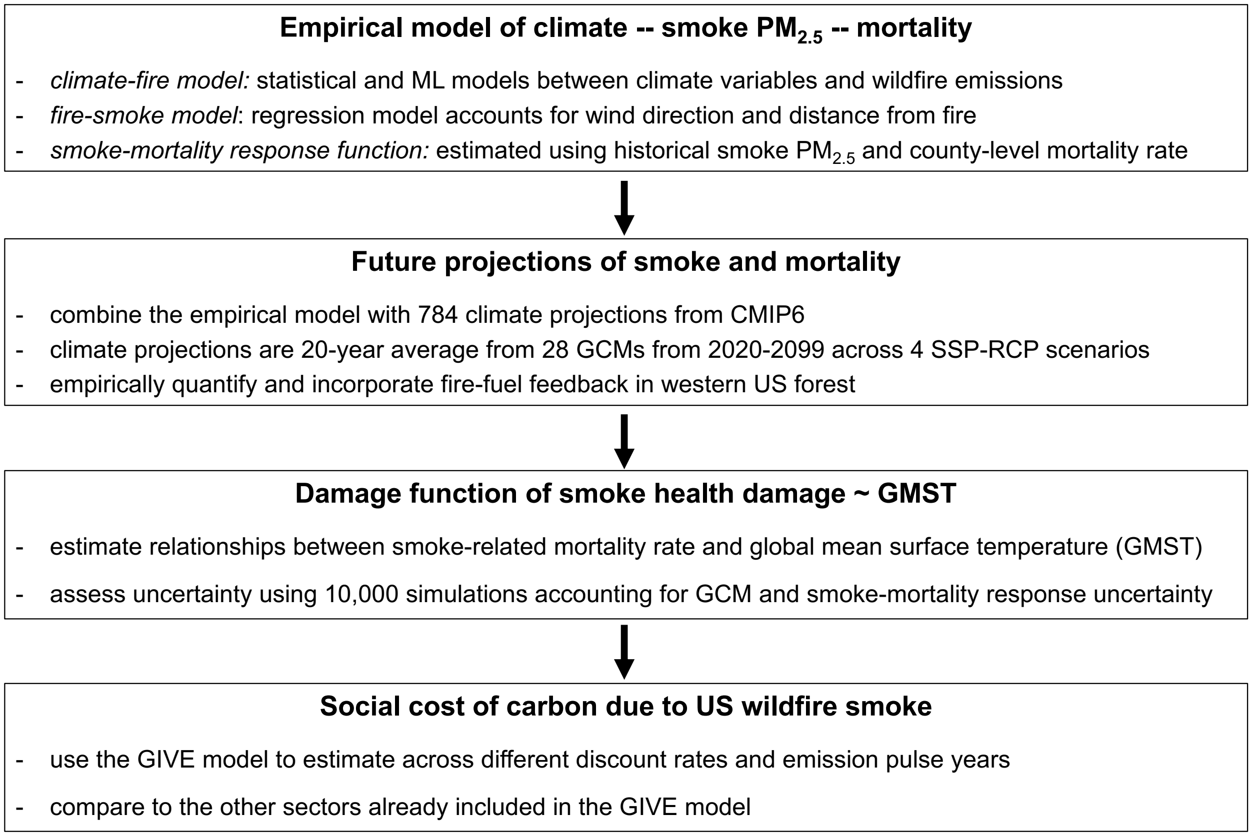


Figure S1: An overview of our research method.

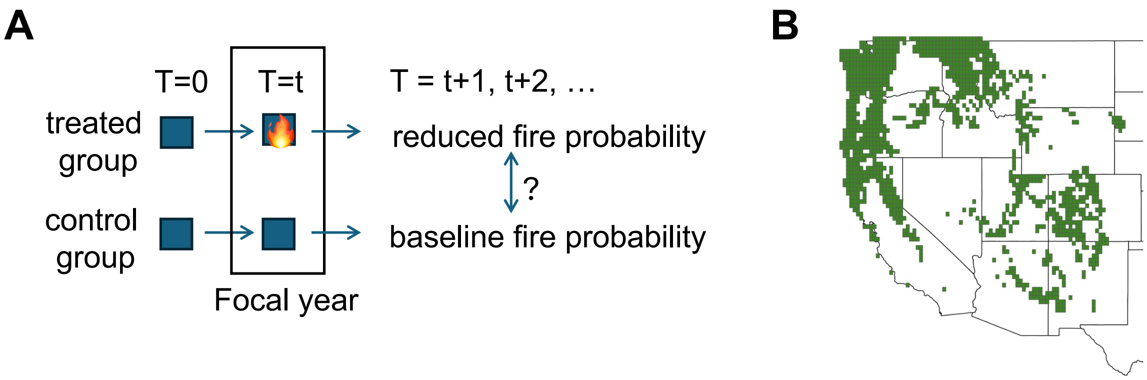


Figure S2: Panel A illustrates our empirical method to quantify the fire-fuel feedback. Panel B shows the forest grid cells in the western U.S. where we quantify the fire-fuel feedback and the influence of fire-fuel feedback on projected emissions.

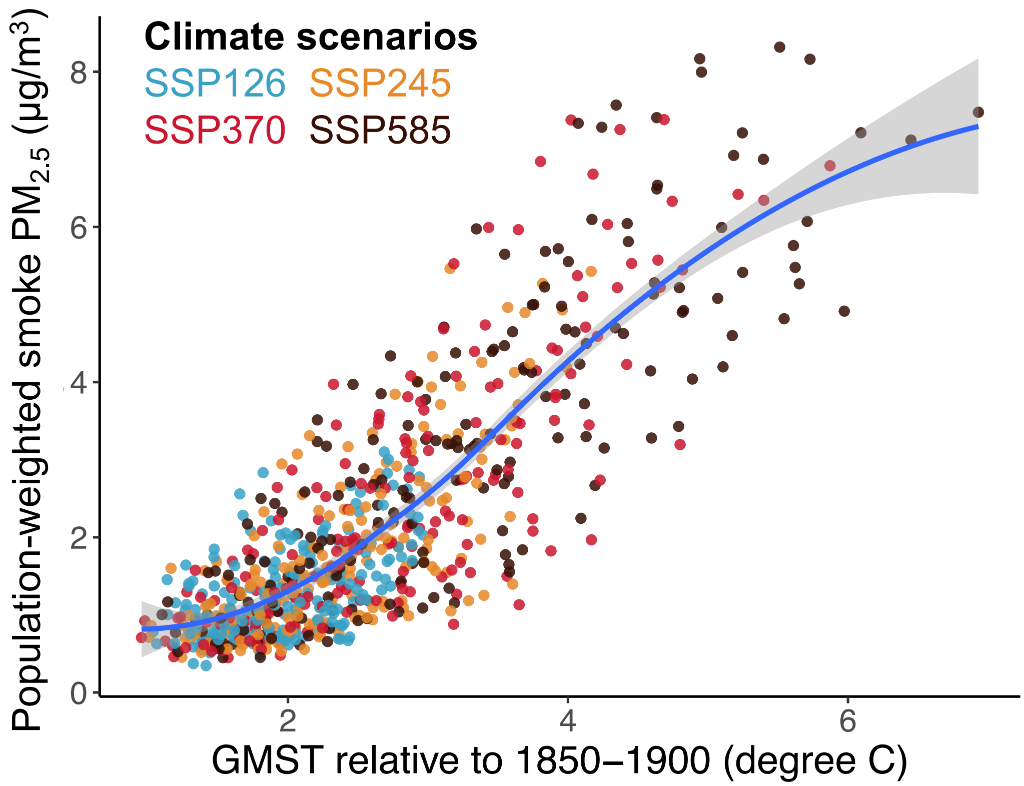


Figure S3: Annual population-weighted smoke $PM_{2.5}$ concentrations across different GMST levels. The population distribution is held constant at the historical level (year 2019). Colors show the different climate scenarios. The smoothing line is fitted using the loess method.

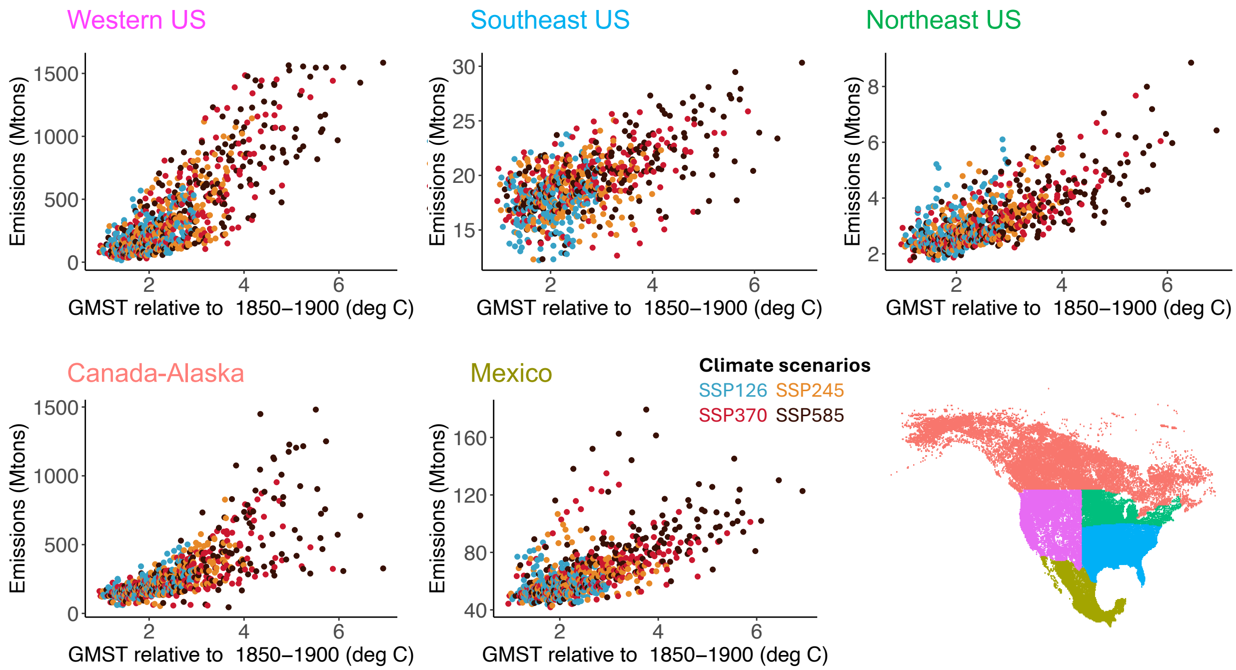


Figure S4: Projected wildfire Dry Matter (DM) emissions across different GMST aggregated at the regional level. We build separate models to predict wildfire DM emissions for five regions, respectively: Western U.S., Southeast U.S., Northeast U.S., Canada-Alaska, and Mexico.

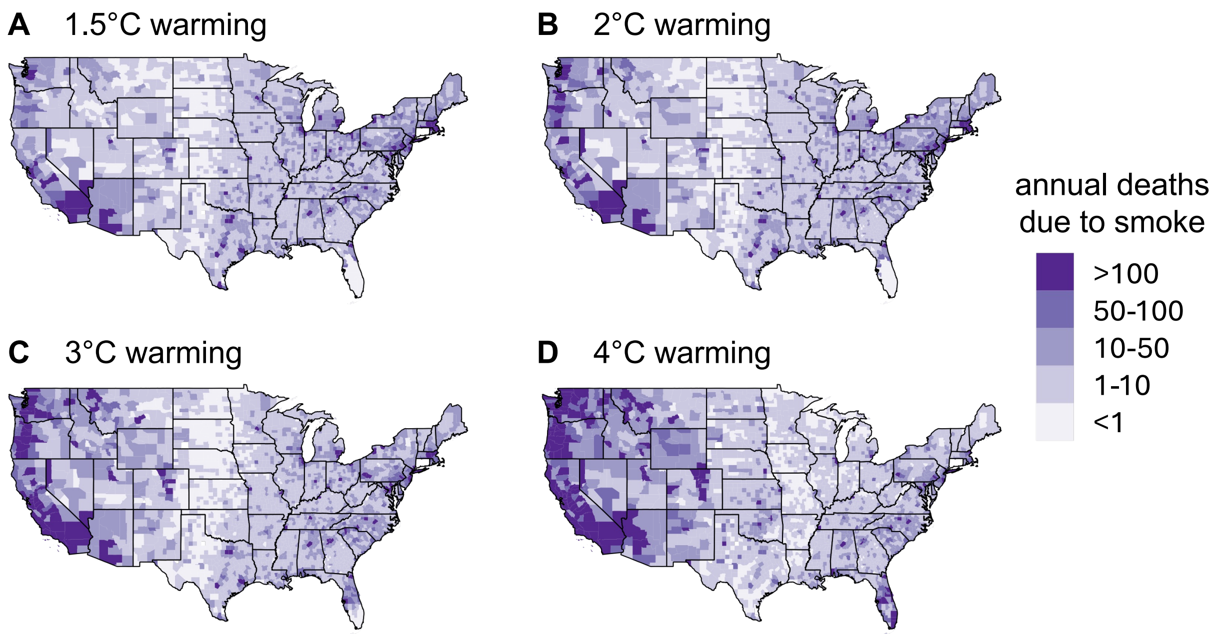


Figure S5: Projected annual mortality due to wildfire smoke $PM_{2.5}$ exposure at the county level. For each warming scenario, the plot shows the median across climate models.

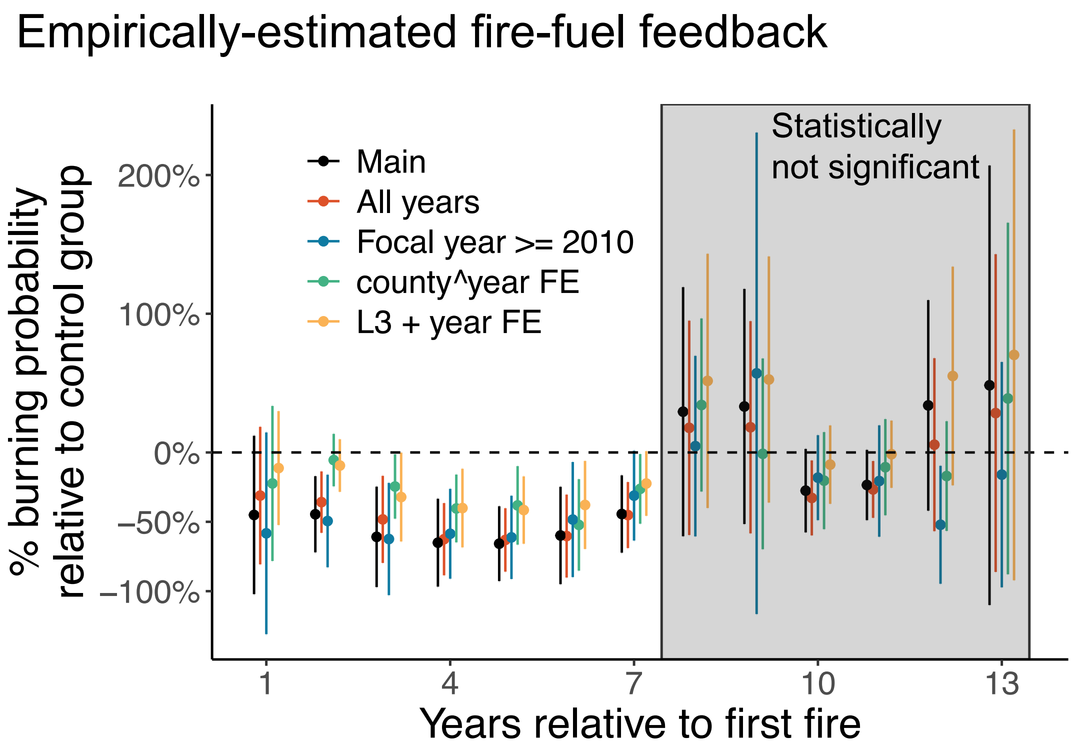
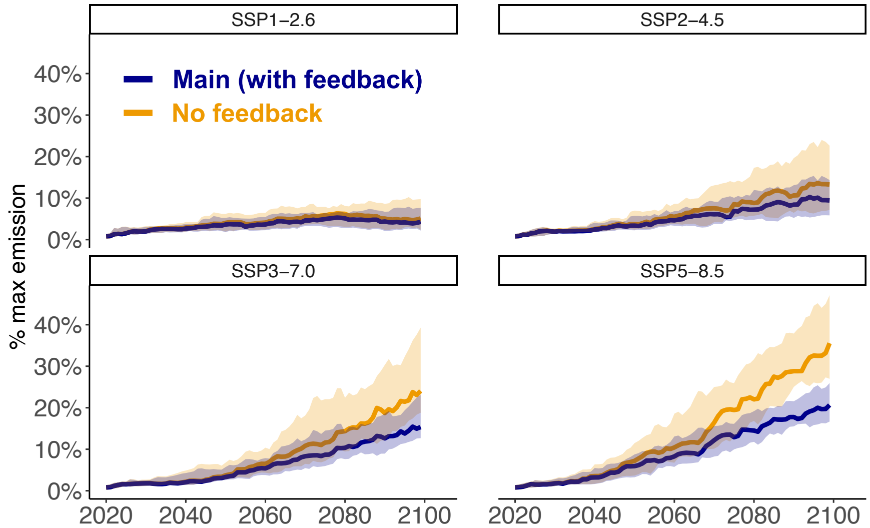


Figure S6: Percentage changes in burning probability in a pixel after X years of fire (shown in the x-axis), relative to pixels that are in the control group. "Main" shows our main estimate reported in the main text using the county and year fixed effects and excluding data with focal years < 2005. "All years" shows results estimated using all data across all focal years. "Focal year $\geq$ 2010" shows results estimated data with only focal years $\geq$ 2010. "county^year FE" shows results estimated using county-year fixed effects. "L3+year FE" shows results estimated using level-III ecoregion and year fixed effects.



Relative difference between feedback and no-feed back scenarios

	2050	2100
SSP1-2.6	-9.9%	-16.0%
SSP2-4.5	-9.8%	-28.9%
SSP3-7.0	-11.0%	-36.2%
SSP5-8.5	-17.7%	-41.9%

Figure S7: Projected fire emissions in the western U.S. forested region (as a percentage of the estimated max emissions) with and without the fire-fuel feedback. The table shows the relative difference between the two scenarios = ("main" - "no feedback") / "no feedback". Our main estimate which accounts for the fire-fuel feedback in western U.S. is shown in blue, and the scenario without the fire-fuel feedback is shown in orange.

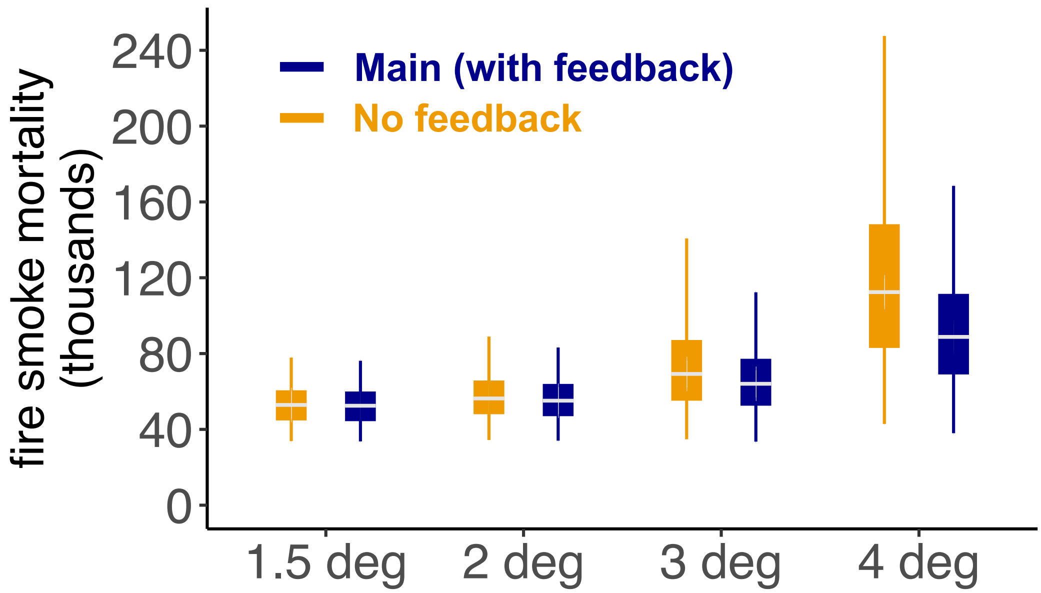


Figure S8: Estimated annual mortality due to wildfire smoke $PM_{2.5}$ across different GMST levels. Our main estimate, which accounts for the fire-fuel feedback in the western U.S., is shown in blue, and the scenario without the fire-fuel feedback is shown in orange.

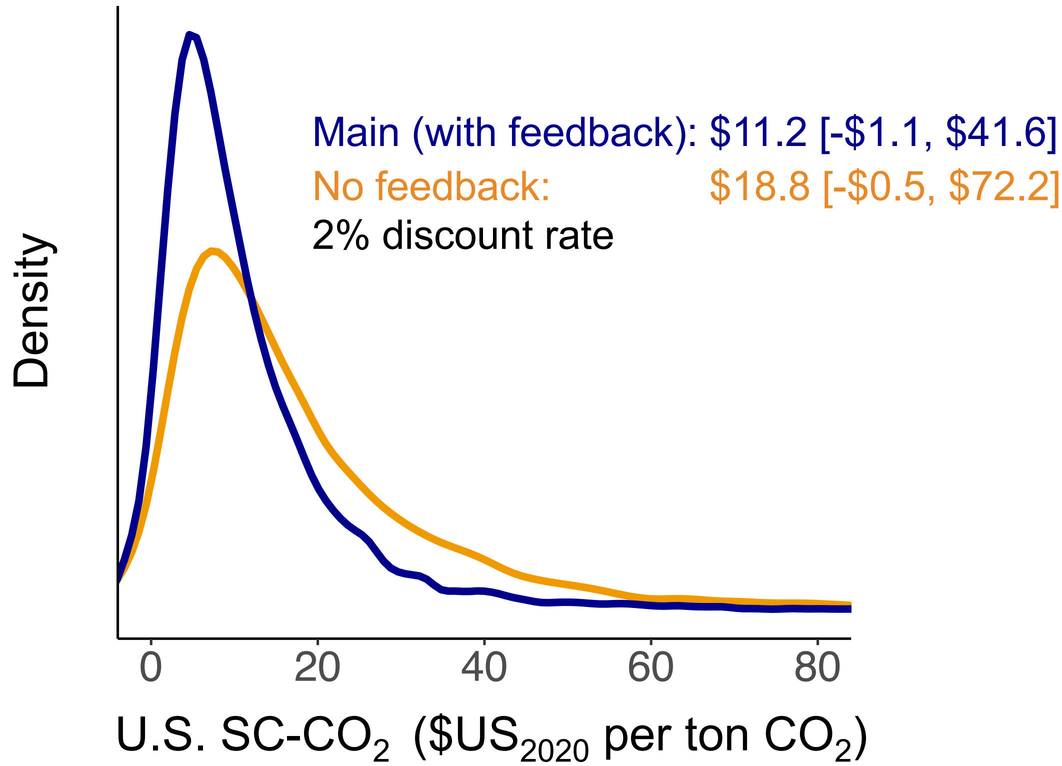


Figure S9: Estimated U.S. SC-CO_2 with and without accounting for the fire-fuel feedback in the western U.S. (2% near-term discount rate). Our main estimate accounts for the fire-fuel feedback in the western U.S. The mean value and 95% confidence interval (2.5 and 97.5th percentile) are shown in the text.

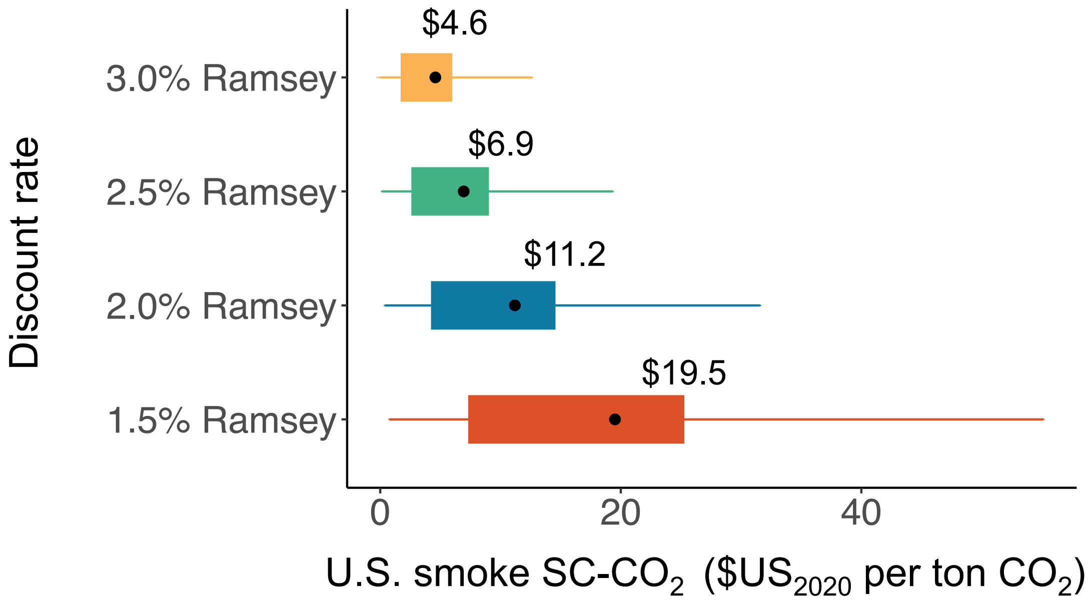


Figure S10: Estimated U.S. SC-CO_2 due to wildfire smoke mortality across different discount rates. The dot shows the mean value, the solid bar shows the 25th and 75th percentiles, and the line shows the 10th and 90th percentiles. Units of U.S. SC-CO_2 are U.S. dollars in the year 2020 per ton of CO_2.

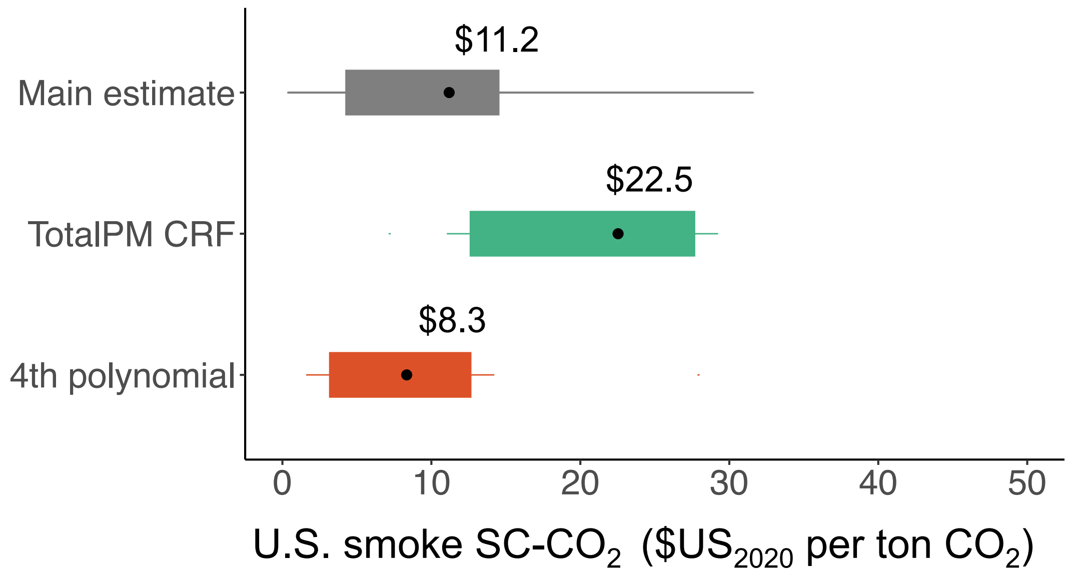


Figure S11: Estimated U.S. SC-CO_2 due to wildfire smoke mortality across alternative model specifications. All results are estimated using a 2% discount rate. "Main estimate" shows our main estimate reported in the main text using the two-part exposure-response function between smoke and mortality and the quadratic model between damage and GMST. "TotalPM CRF" shows results estimated using an exposure-response function derived from all-source total $PM_{2.5}$ from Pope et al., 2020. "4th polynomial" shows results estimated using 4th-order polynomial damage functions between smoke damage and GMST. The dot shows the mean value, the solid bar shows the 25th and 75th percentiles, and the line shows the 10th and 90th percentiles. Units of U.S. SC-CO_2 are U.S. dollars in the year 2020 per ton of CO_2.

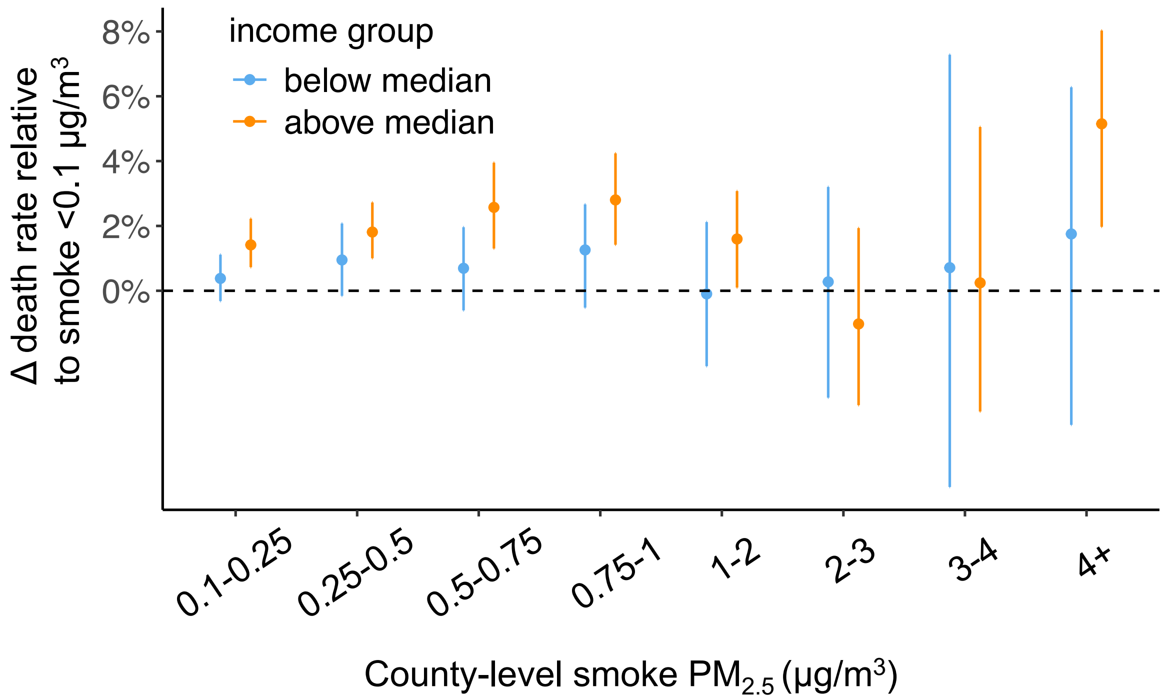


Figure S12: Estimated smoke-mortality exposure-response functions across different income groups. We split all counties in our sample into two groups based on the average county income (below or above the median) and separately estimate the exposure-response function for each group. The plot shows effects of exposure to different annual mean concentrations of smoke $PM_{2.5}$ (x-axis) relative to a year with smoke concentration <0.1 $\mu g/m^3$. The error bars show 95% confidence intervals. County income data are derived from the U.S. Census 2017-2021 American Community Survey.

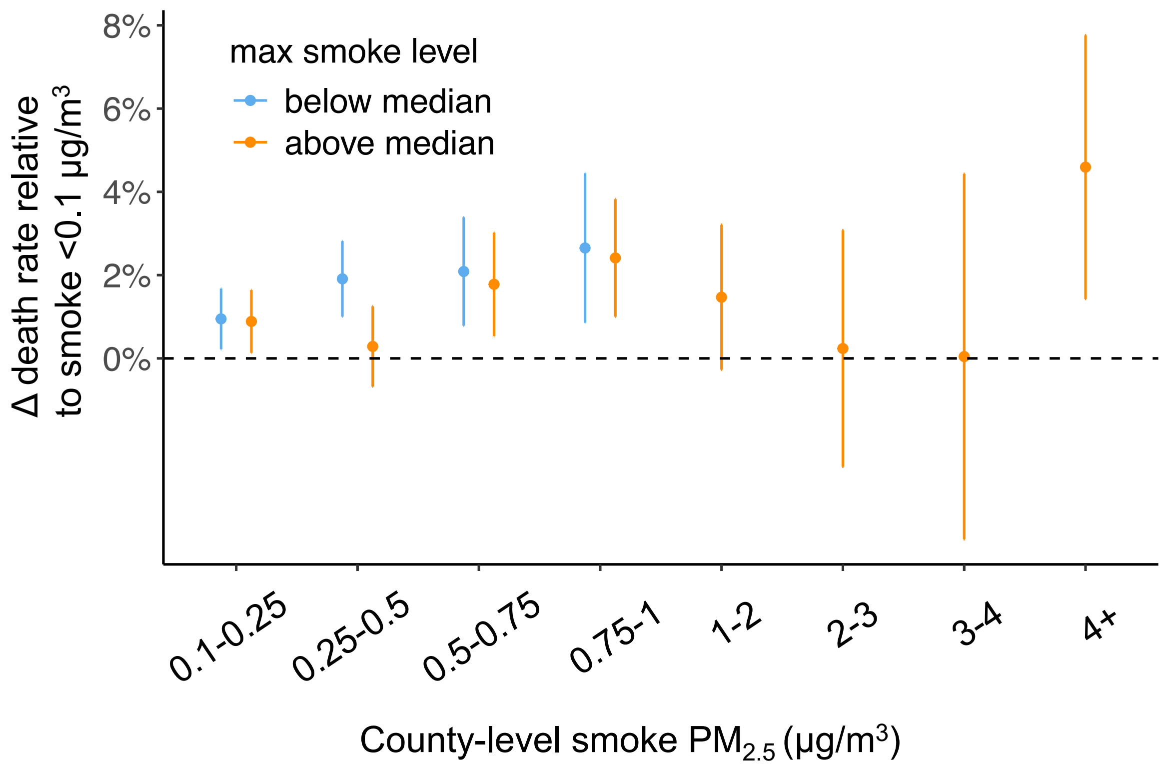


Figure S13: Estimated smoke-mortality exposure-response functions across different baseline smoke levels. We split all counties in our sample into two groups based on their maximum annual smoke levels (below or above the median) and separately estimate exposure-response function for each group. The plot shows effects of exposure to different annual mean concentrations of smoke $PM_{2.5}$ (x-axis) relative to a year with smoke concentration <0.1 $\mu g/m^3$. The error bars show 95% confidence intervals. Note that there are no estimated coefficients for some bins in the low-smoke group because this group has not experienced smoke levels in those ranges.

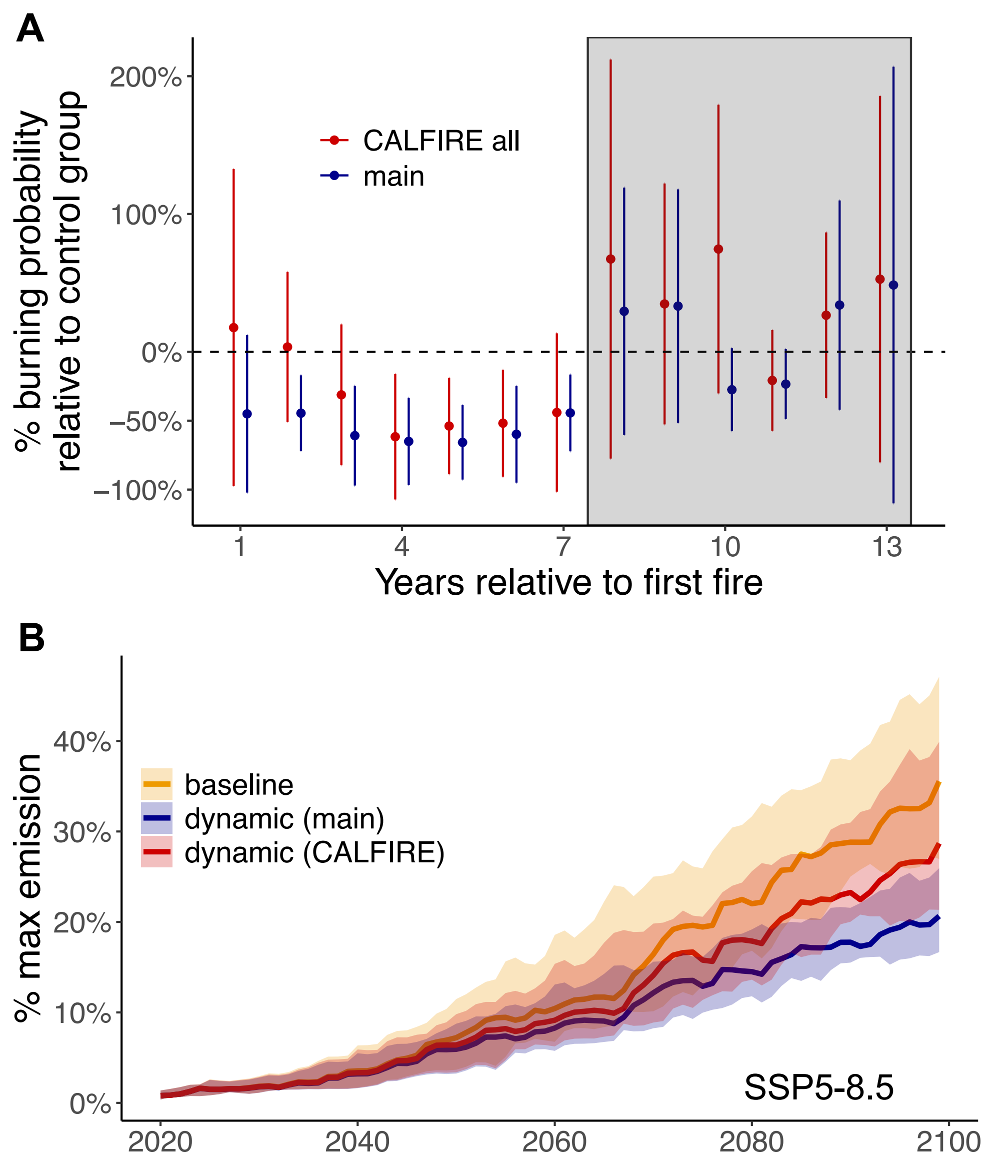


Figure S14: Estimated fire-fuel feedback using a more restricted sample. Panel A: Percentage changes in burning probability in a pixel after X years of fire (shown in the x-axis), relative to pixels that are in the control group. "Main" shows our main estimate reported in the main text using the full sample (all pixels that burned 0-2 times) with focal years > 2005. "CALFIRE all" shows results estimated using all pixels with fire risks determined as "moderate", "high", or "very high", regardless of the number of fires happened in the sample. The fire risk is derived from CALFIRE fire hazard zones (*80*). Thus "CALFIRE all" shows results estimated using a more restricted sample, which only included pixels with some level of fire risk. Panel B: Projected fire emissions in the western U.S. forested region (as a percentage of the estimated max emissions) using different estimated feedback. Our main estimate which accounts for the fire-fuel feedback in western U.S. is shown in blue, the scenario using feedback estimated using alternative sample is shown in red, and the scenario without the fire-fuel feedback is shown in orange.

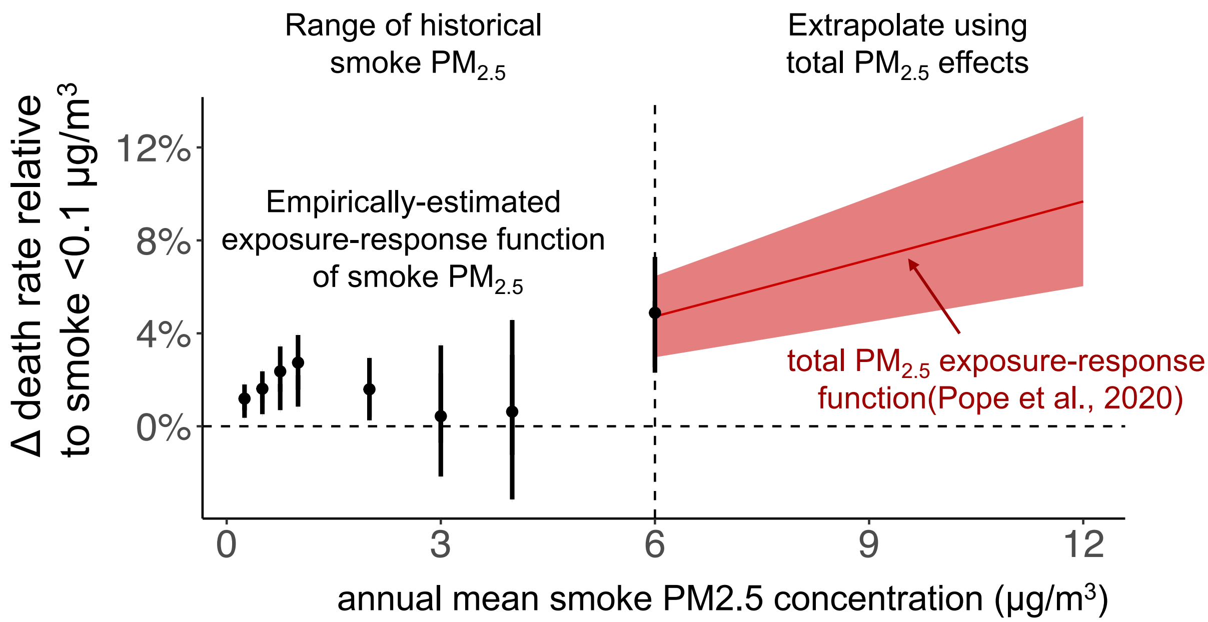


Figure S15: The two-part exposure-response function between smoke $PM_{2.5}$ concentrations and all-cause mortality. The black dots represent the empirically estimated exposure-response function linking smoke concentrations to all-cause mortality. We use this exposure-response function to calculate the smoke-related mortality for all county-years with an annual mean smoke $PM_{2.5}$ concentration below 6 $\mu g/m^3$. The red line shows the exposure-response function for all-source total $PM_{2.5}$ derived from Pope et al., 2020 (*81*). We use this part to calculate smoke-related mortality for any counties with annual mean concentration above 6 $\mu g/m^3$. The error bars of the black dots denote the 95% confidence interval estimated by bootstrapping. The shaded area of the red line represents the 95% confidence intervals as reported in Pope et al., 2020.

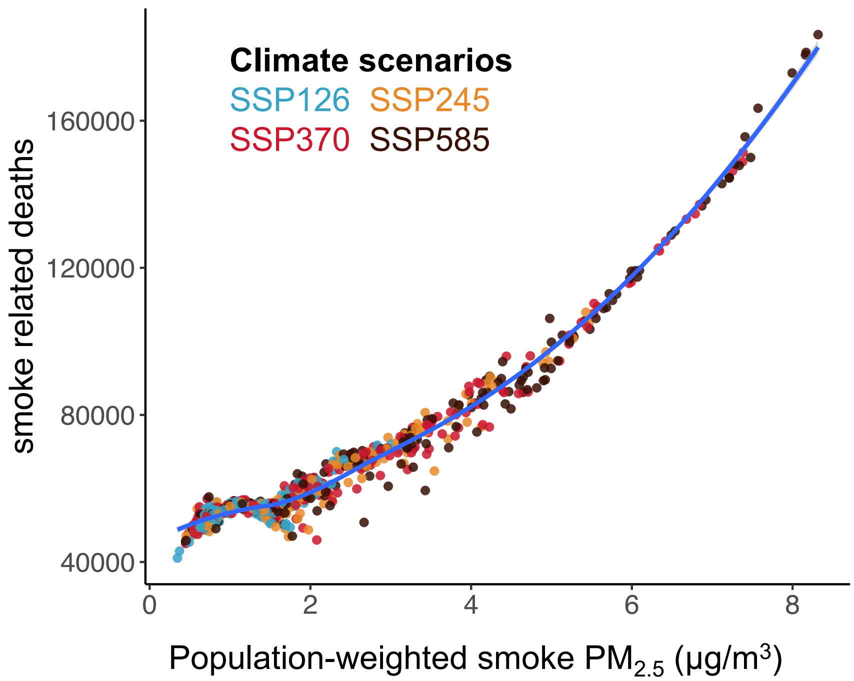


Figure S16: Estimated smoke-related mortality across different levels of population-weighted smoke $PM_{2.5}$ concentrations. Population distribution and density are fixed at the 2019 level. The smoothing line is fitted using the loess method.

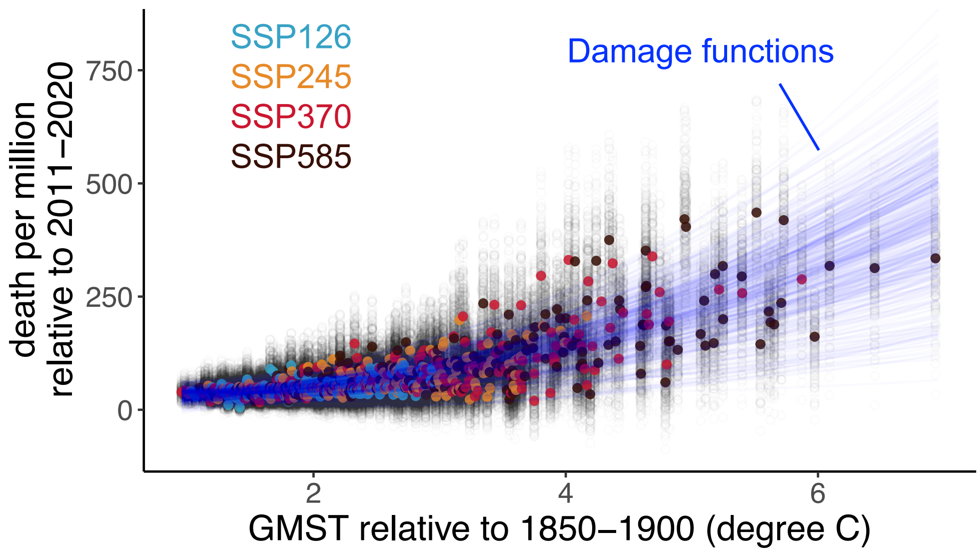


Figure S17: Estimated damage functions between wildfire smoke death rate (unit: death per million) and GMST. The blue line shows the damage functions estimated using bootstrap and a quadratic model. We assess the uncertainty of damage functions using a Monte-Carlo simulation of 10,000 runs (only 500 are shown in the figure for visualization purposes). Each colored dot represents smoke-related death rate for one GCM and scenario using the main health exposure-response function. Grey dots represent smoke-related death estimated using alternative health exposure-response functions and thus represent the uncertainty from the exposure-response function.